ASSIGNMENT 10

NAME: BUDDHABHUSHAN SIRSAT
ROLL NO.: 99421

1.

```sql
CREATE TABLE RES(L INT, W INT, PERIMETER DOUBLE, AREA DOUBLE);

DELIMITER $$
CREATE PROCEDURE P(IN p_L INT, IN p_W INT)
BEGIN
 INSERT INTO RES VALUES(p_L, p_W, 2*(p_L+p_W), p_L*p_W);
END $$
DELIMITER ;

  CALL P(4,5);

  SELECT * FROM RES;
```

OUTPUT:

```
+-------+-------+-----------+-------+
| L     | W     | PERIMETER | AREA  |
+-------+-------+-----------+-------+
|     4 |     5 |        18 |    20 |
+-------+-------+-----------+-------+
```

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2.

```
CREATE TABLE TEMP(num INT, SQUARE INT, CUBE_OF_NUM INT);

DELIMITER $$
CREATE PROCEDURE P()
BEGIN
 DECLARE num INT DEFAULT 2;
 INSERT INTO TEMP VALUES(num, num*num, num*num*num);
END $$
DELIMITER ;

CALL P();
SELECT * FROM TEMP;
```

OUTPUT:

```
+------+--------+-------------+
| num  | SQUARE | CUBE_OF_NUM |
+------+--------+-------------+
|    2 |      4 |           8 |
+------+--------+-------------+
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3.

```sql
DELIMITER $$
CREATE PROCEDURE TempConvert(IN val DOUBLE, IN unit CHAR(1))
BEGIN
 IF unit = 'F' THEN
   SELECT val AS Fahrenheit, (val - 32) * 5/9 AS Celsius;
 ELSEIF unit = 'C' THEN
   SELECT val AS Celsius, (9/5 * val) + 32 AS Fahrenheit;
 ELSE
   SELECT 'Invalid unit. Use F or C' AS Message;
 END IF;
END $$
DELIMITER ;


CALL TempConvert(98.6, 'F');
CALL TempConvert(37, 'C');
```

OUTPUT:

```
+------------+---------+
| Fahrenheit | Celsius |
+------------+---------+
|       98.6 |      37 |
+------------+---------+
+---------+-------------------+
| Celsius | Fahrenheit        |
+---------+-------------------+
|      37 | 98.60000000000001 |
+---------+-------------------+
```

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4.

```
DELIMITER $$
CREATE PROCEDURE InchConvert(IN total_inches INT)
BEGIN
 DECLARE yards INT;
 DECLARE feet INT;
 DECLARE inches INT;
  SET yards = total_inches / 36;
 SET feet = (total_inches % 36) / 12;
 SET inches = total_inches % 12;
  SELECT total_inches AS Total_Inches, yards AS Yards, feet AS Feet,
inches AS Inches;
END $$
DELIMITER ;


CALL InchConvert(124);
```

OUTPUT:

```
+--------------+-------+------+--------+
| Total_Inches | Yards | Feet | Inches |
+--------------+-------+------+--------+
|          124 |     3 |    1 |      4 |
+--------------+-------+------+--------+
```

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5.

```
DELIMITER $$
CREATE PROCEDURE ProductCheck(IN num1 DOUBLE, IN num2 DOUBLE)
BEGIN
 DECLARE prod DOUBLE;
 SET prod = num1 * num2;
  IF prod >= 100 THEN
   SELECT prod AS Product, 'Greater than or equal to 100' AS Result;
 ELSE
   SELECT prod AS Product, 'Less than 100' AS Result;
 END IF;
END $$
DELIMITER ;


CALL ProductCheck(12.5, 9);
CALL ProductCheck(5, 6);
```

OUTPUT:

```
+---------+------------------------------+
| Product | Result                       |
+---------+------------------------------+
|   112.5 | Greater than or equal to 100 |
+---------+------------------------------+
+---------+---------------+
| Product | Result        |
+---------+---------------+
|      30 | Less than 100 |
+---------+---------------+
```